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Genome Note

# Epidemiological pattern and genomic insights into multidrug-resistant ST491 *Acinetobacter baumannii* BD20 isolated from an infected wound in Bangladesh: Concerning co-occurrence of three classes of beta lactamase genes

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ABSTRACT

*Objective:* Multidrug-resistant (MDR) *Acinetobacter baumannii* is a major issue within healthcare facilities in Bangladesh due to its frequent association with hospital-acquired infections. In this study we report on a carbapenem-resistant draft genome sequence of an *A. baumannii* BD20 sample isolated from an infected wound in Bangladesh.
*Methods:* *A. baumannii* BD20 was isolated from an infected burn wound. Whole-genome sequencing was carried out and annotated using PGAP and Prokka. Sequence type, antimicrobial resistance genes, virulence factor genes, and metal resistance genes were investigated. Core genome multilocus sequence typing–based phylogenomic analysis between *A. baumannii* BD20 and 213 *A. baumannii* strains retrieved from the NCBI GenBank database was performed using the BacWGSTdb 2.0 server.
*Results:* *A. baumannii* BD20 (MLST 491) was resistant to all the antibiotics tested, except for colistin and polymyxin B. Along with many other antibiotic resistance genes, the isolate harbored three classes of beta lactamase–producing genes: $bla_{\text{GES-11}}$ (class A), $bla_{\text{OXA-69}}$ (class D), $bla_{\text{ADC-10}}$ (class C), and $bla_{\text{ADC-11}}$ (class C). Additionally, the strain carried several virulence genes and metal resistance determinants, which may contribute to its increased virulence. Core genome MLST–based phylogenomic analysis revealed that *A. baumannii* BD20 was closely related to another ST491 strain isolated from Singapore.
*Conclusions:* The findings of this study underscore the growing challenge of MDR *A. baumannii*, emphasizing the need for vigilant surveillance and infection-control measures in healthcare settings in order to address these emerging threats effectively.



## 1. Introduction

The incidence of multidrug-resistant (MDR) bacterial strains, especially those resistant to carbapenems, is steadily increasing in Bangladesh, creating a concerning situation [1–3]. This growing public health threat is characterized by bacteria that can produce enzymes, such as carbapenemases, which neutralize carbapenems that are often used as a last resort for treating severe bacterial infections, particularly those caused by MDR organisms [4–6]. Several studies have been conducted to examine the patterns of multidrug resistance and the molecular epidemiology of pathogens prevalent in Bangladesh, particularly within healthcare facilities [5,7–9]. Among non-Enterobacteriaceae microorganisms, *Acinetobacter baumannii* has become a significant concern in contemporary healthcare, presenting as a Gram-negative nosocomial pathogen with extensive resistance to the majority of existing antibiotics [10].

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MDR *A. baumannii* (MDR-AB) stands as a formidable challenge within contemporary healthcare systems. This pathogen's ability to resist multiple antibiotics, including the critically important carbapenems, has placed it at the forefront of concerns in clinical settings worldwide [11]. Beyond its intrinsic resilience, the epidemic sequence types associated with MDR-AB have garnered attention due to their propensity for rapid dissemination, amplifying the pathogen's impact [12]. Among MDR-AB, carbapenem-resistant *A. baumannii* is a major public health challenge, particularly in healthcare settings with high antibiotic misuse and inadequate infection control, such as those found in Bangladesh [13]. Carbapenem-resistant *A. baumannii* exhibits resistance through mechanisms such as carbapenemase production, efflux pumps, and biofilm formation, complicating treatment options [14]. The bacterium's persistence on surfaces and resistance to disinfectants makes it a frequent cause of hospital outbreaks, particularly in intensive care units. Moreover, MDR-AB's capacity to maintain virulence factors further complicates the clinical landscape, necessitating a comprehensive understanding of both its drug-resistance mechanisms and epidemiological characteristics for effective management and infection-control strategies.

*A. baumannii* is a significant concern in healthcare settings in Bangladesh, as it is often associated with hospital-acquired infections and exhibits varying levels of antibiotic resistance, posing challenges for effective treatment and infection-control measures [15]. In this study we have revealed the genomic characteristics of a clinical strain of MDR *A. baumannii* isolated from Bangladesh.

## 2. Materials and methods

A wound swab sample was collected from Holy Family Red Crescent Hospital (Dhaka, Bangladesh). The sample was immediately transported to the Microbiological Laboratory within a cooler with icepacks (below 4 °C) and cultured on MacConkey agar and Mueller Hinton agar plates. Predominant similar isolated colonies were selected and a single colony was sub-cultured to obtain a pure colony; that isolate was subjected for further analysis. The isolate was screened phenotypically for resistance to 14 antibiotics from eight different classes using the Kirby-Bauer disk diffusion method [16] in Mueller Hinton agar media and by determining the minimum inhibitory concentration (MIC) according to Clinical and Laboratory Standard Institute guidelines [17]. The 14 antibiotics examined were as follows: ampicillin-sublactam, doxycycline, gentamicin, amikacin, trimethoprim-sulfamethoxazole, tetracycline, imipenem, meropenem, doripenem, ceftazidime, ciprofloxacin, cefotaxime, polymyxin B, and colistin (Oxoid Ltd., Basingstoke, UK). MIC breakpoints were determined for colistin, polymyxin B, meropenem, imipenem, and tetracycline. A biofilm-formation assay was performed in 96-well flat-bottomed wells, polystyrene plate (Costar, Corning, NY, USA), following the standard protocol of a crystal violet assay [18]. Isolate genomic DNA was extracted from overnight-grown culture using a Monarch Genomic DNA Purification Kit (New England Biolabs, MA, USA) according to the manufacturer's instructions. Whole-genome sequencing of the target isolate was performed under a Illumina platform using the Illumina MiniSeq Sequencing System (CA, USA) at the Genome Research Laboratory, Bangladesh Council of Scientific and Industrial Research (Dhaka, Bangladesh).

The generated FASTQ files were evaluated for quality using FastQC (v0.11) [19]. Adapter sequences and low-quality ends per reading were trimmed using Trimmomatic (v0.39) [20], and high-quality reads underwent de novo assembly using SPAdes (Species Prediction and Diversity Estimation) v3.13 [21]. The assembled genome was identified using the KmerFinder 3.1 tool (https://cge.food.dtu.dk/services/KmerFinder/). Sequence typing of the genome was carried out using PubMLST (https://pubmlst.org/organisms). We used the NCBI Prokaryotic Genome Annotation Pipeline (PGAPv4.11) with a best-placed reference protein set and Prokka [22] for genome annotation. A graphical map of the circular genome was generated using the Proksee server (https://proksee.ca/).

Antimicrobial resistance genes (ARGs) were investigated through Resistance Gene Identifier (RGI) of the Comprehensive Antibiotic Resistance Database (CARD) (https://card.mcmaster.ca/analyze/rgi). Along with that, metal resistance genes (MRG) were further investigated using BacMet [23], a tool for the identification of biocide and metal resistance genes. Virulence factor genes were investigated through VFDB (https://www.mgc.ac.cn/cgi-bin/VFs/v5/main.cgi). Mobile genetic elements were detected through the MobileElementFinder tool (https://cge.food.dtu.dk/services/MobileElementFinder/) and mobileOG-db [24]. Phage integration inside the bacterial genome was inspected using the Phigaro [25] and Virsorter [26] tools. In order to understand the genomic epidemiology, we used BacWGSTdb 2.0 (https://bacdb.cn/BacWGSTdb), which uses both core genome multilocus sequence typing and a single-nucleotide polymorphism–based scheme for genomic comparison.

## 3. Results and discussion

The test isolate was identified as a strain of *A. baumanii* based on its 16S rDNA and whole-genome sequence, and the strain was further identified as *A. baumanii* BD20. The isolate was found to be completely resistant to all the tested antibiotics except for colistin and polymyxin B (Supplementary Table 3). The MICs of the isolates for imipenem, meropenem, chloramphenicol, and tetracycline were 256, >256, >512, and >512 mg/L, respectively. The draft genome length was 4,023,613 bp, having 58 contigs with a GC content of 38.89%. The CDS (Coding Sequence) ratio of 0.972 is indicative of a meticulous assembly procedure, aligning closely with the standard CDS ratio range recommended by NCBI, which typically spans from 0.8 to 1.2 for a conventional assembly. From CARD:RGI analysis, it was revealed that the isolate harbored multiple ARGs, including several efflux pump genes and beta lactamase genes (Supplementary Table 1), and showed the co-existence of $bla_{\text{GES-11}}$ (class A), $bla_{\text{OXA-69}}$ (class D), $bla_{\text{ADC-10}}$ (class C), and $bla_{\text{ADC-11}}$ (class C) genes in its genome (Fig. 1C and D). The resistance mechanism displayed by the isolate against carbapenem antibiotics such as imipenem and meropenem can be attributed to the presence of the $bla_{\text{GES-11}}$ gene. This is consistent with prior findings that have associated $bla_{\text{GES}}$ genes with the development of resistance to carbapenem antibiotics [27]. Additionally, it is worth noting that the concurrent existence of two closely related variants of class C beta lactamase genes, such as $bla_{\text{ADC-10}}$ and $bla_{\text{ADC-11}}$, is an infrequent phenomenon within the context of any bacterial isolate. In order to confirm the variants, the positions of the genes were inspected and both of the gene sequences were aligned with the reference sequences. These significant sequence similarities with their respective reference sequences and distinctive locations in the genome confirmed the identification of $bla_{\text{ADC-10}}$ and $bla_{\text{ADC-11}}$ by eliminating the possibility of sequencing bias. The presence of multiple antibiotic resistance determinants assembled within the genome could be characterized as both genomic and plasmid mediated. Interestingly, resistome and variant analysis revealed several acquired ARGs, including beta lactamase.

In addition to beta lactamases, the strain BD20 harbored a wide array of other antibiotic resistance genes including *aph(3′)-Ia, aac(6′)-Ib3, aph(3′)-VIa, ant(3′′)-llc, aac(3)-la*, and *aad*A, which are responsible for conferring resistance to aminoglycosides. The *aph(3′)-Ia* gene is responsible for conferring resistance to streptomycin, while *aph(3′)-VIa* confers resistance to amikacin and kanamycin. Additionally, *ant(3′′)*-llc is associated with resistance to

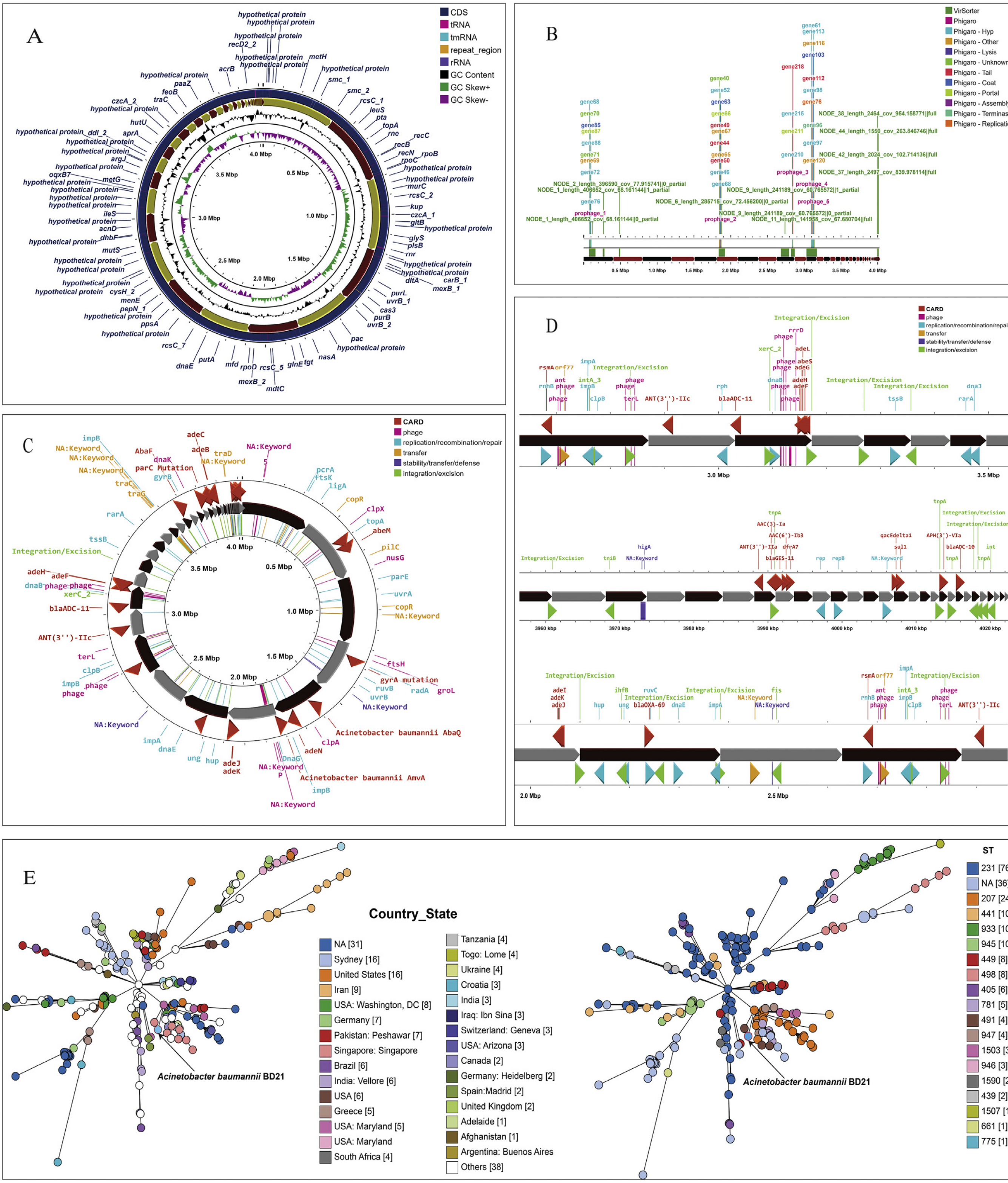


**Fig. 1.** Genomic characterization of *Acinetobacter baumannii* BD20. (A) Genome mapping of *A. baumannii* BD20. (B) Prophage integration within the genome. (C) Mobile genetic elements and antimicrobial resistance gene mapping. (D) Investigation of mobile genetic elements around major antimicrobial resistance genes. (E) Phylogenomic relationship of *A. baumannii* BD20 with 213 other *A. baumannii* strains deposited in the NCBI Genbank database. Lines connecting the circles indicate the clonal relationship between different isolates.

both streptomycin and spectinomycin in *A. baumannii*. Strain BD20 also harbored resistance genes against sulfonamide (*sul*1) and trimethoprim (*dfr*A7). Furthermore, the isolate also possessed an array of multiple efflux pumps conferring resistance to macrolide, fluoroquinolone, cephalosporin, carbapenem, and tetracycline; the simultaneous presence of different types of efflux pumps increases resistance to several antimicrobials. Additionally, target alterations in proteins GyrA (S81L) and ParC (S84L, V104I, and D105E) were predicted to be responsible for conferring resistance to fluoroquinolones. These findings accurately explain the MDR characteristics seen against aminoglycosides, carbapenems, quinolones, cephalosporins, and tetracyclines for this isolate. Mobile genetic element analysis revealed the presence of genes such as *ihf*B, *xer*C, *int*A, *tnp*A, and *tni*B, as well as many other genes around the ARGs that are involved in excision, integration, recombination, and transfer of genetic materials. Additionally, insertion sequences, composite transposons, and unit transposon-type mobile genetic elements such as IS26, cn_2249_IS26, ISAba26, Tn6018, ISAba125, and ISAba13 were detected after mining the genome sequence of *A. baumannii* BD20. This suggested an alarming level of horizontal transfer of ARGs and their dissemination. Prophage investigation revealed the integration of sequences of *Myoviridae* and *Siphoviridae* within the genome (Fig. 1B).

Genotypic and phenotypic analysis revealed important insights into virulence and metal resistance. In our study, a metal-sensitivity assay suggested that this isolate is resistant to several metal ions including copper, zinc, iron chromium, cobalt, and silver. The result of the metal-sensitivity test showed concordance with the genotypic resistance profiles. The isolate possessed an arsenal of metal resistance determinants including *nre*B, *czc*A, *czc*D, *mgt*A, *cop*R, *mgt*A, *rec*G, *ruv*B, *irl*R, *gol*T, *ars*B, *ars*C, *chr*A, *dme*F, *mod*B, and so on (Supplementary Table 2), which are both chromosomal and plasmid mediated and thus may confer resistance to a wide range of heavy metal ions. Such extensive metal resistance within this isolate may contribute to increased antimicrobial resistance. It is also evident that the co-occurrence ARG and MRG is more frequent in pathogenic than in nonpathogenic isolates, thus posing an increased threat to human health, which echoes several studies that have reported that ARG and MRG co-occurrence is significantly more frequent in pathogenic genomes compared with nonpathogenic genomes. Virulence gene investigation revealed the presence of several genes involved in biofilm formation (*ade*F, *csu*A, *csu*B, *pga*A, *pga*B, *pga*C, and *pga*D), adherence (*omp*A), enzymes (*plc*C and *plc*D), immune evasion (*lps*B, *lpx*A, *lpx*B, and *lpx*C), quorum sensing (*aba*I and *aba*R), and serum resistance (*pbp*G). The co-occurrence of virulence factor genes, and metal and antibiotic resistance determinants, may pose an increased threat to human health.

According to in silico MLST analysis, *A. baumannii* BD20 belongs to sequence type 491 (ST491). In order to examine the genomic epidemiological attributes of *A. baumannii* strains on a global scale, an analysis was conducted to assess the phylogenetic linkage between *A. baumannii* BD20 and a comprehensive set of 213 *A. baumannii* strains that are presently archived within the NCBI GenBank database (Fig. 1E). Core genome multilocus sequence typing–based phylogenomic analysis revealed that *A. baumannii* BD20 was closely related to another ST491 strain isolated from Singapore. This strain also harbors the $bla_{OXA-69}$ gene, similar to the BD20 strain. Throughout this analysis, it was revealed that ST231 has been the dominant strain worldwide, followed by ST207, ST441, ST933, and ST945.

## 4. Conclusion

In conclusion, we have reported on the genomic characteristics of MDR ST491 *A. baumannii* strain BD20 isolated from an infected wound swab of a patient in Bangladesh. The co-existence of class A, C, and D beta lactamase genes, including two close variants of $bla_{ADC}$- and $bla_{GES}$-like carbapenemase genes associated within mobile genetic elements, indicates an alarming situation regarding the dissemination of ARGs in both a local and global context. Additionally, the co-existence of metal resistance genes and several major virulence genes emphasizes the noteworthy pathogenic potential of this isolate. Our analysis of the genomic characteristics of MDR-AB BD20, which revealed a range of ARGs, VFGs, and genomic functional potentials, has the potential to provide insights into the mechanisms through which resistance and pathogenic genes are disseminated among bacteria, animals, and humans. These findings pave the way for additional investigations into this pathogen's epidemiology, mechanisms of drug resistance, and genomic features. Sustained vigilance is necessary for the monitoring and prevention of the transmission of MDR *A. baumannii* within clinical settings.

## Data availability

The whole-genome sequence of *Acinetobacter baumannii* BD20 was deposited in GenBank of NCBI under the accession number JAMQYF000000000. The BioProject accession, BioSample accession and Assembly accession numbers are PRJNA846862, SAMN28905563, and GCA_026130565.1, respectively.

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## Declaration of competing interests

None declared.

## Ethical approval

Ethical permission to conduct the study was given and verified by the Ethical Review Committee of Faculty of Biological Sciences, University of Dhaka (ref. no. 191/Biol. Scs.).

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2024.07.009.

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